

A Simple Manual to How Crypto Exchanges Work

This guide explains:

- Market orders & limit orders
- The limit order book (LOB)
- How orders are matched
- Last price, index price, mark/fair price
- Funding rate
- Open interest

Think of this as “what the exchange engine is really doing.”

1. Market Orders vs Limit Orders

Market Order — *“Fill me now”*

- You say: **buy or sell immediately**
- The exchange matches you with the **best available prices** in the order book
- You **don’t control the exact price**
- You pay **taker fees**
- You may get **slippage** if the book is thin or price is moving fast

Example

BTC is trading around 50,000

You place a market buy

You may get fills at 50,001, 50,003, 50,010 — until your size is filled

Limit Order — *“Only at my price or better”*

- You specify a **price**
- Your order **waits in the order book**
- You **control price**, but **may not get filled**
- Often earns **maker fees (or rebates)**

Example

BTC is at 50,000

You place a limit buy at 49,900

Your order sits until someone sells into it

Key Difference (Mental Model)

Order Type	Speed	Price Control	Liquidity
Market	Instant	✗ None	Takes liquidity
Limit	Slower	✓ Full	Provides liquidity

2. The Limit Order Book (LOB)

The **order book** is simply a list of all resting limit orders.

Two sides:

- **Bids** → buy orders
- **Asks** → sell orders

Each price level has:

- Price
- Total size (sum of all orders at that price)

📌 Example snapshot:

Price	Size	Side
50,100	2.5 BTC	Ask
50,050	1.8 BTC	Ask
50,000	—	Last Price
49,950	2.0 BTC	Bid
49,900	3.2 BTC	Bid

Best Bid & Best Ask

- **Best Bid** = highest buy price
- **Best Ask** = lowest sell price
- **Spread** = Best Ask – Best Bid

3. How Orders Are Matched

Crypto exchanges use **price-time priority**:

1. **Best price first**
2. **Earlier orders first**

Matching rules:

- A **market order** consumes limit orders
- A **limit order**:
 - Matches immediately *if it crosses*
 - Otherwise rests in the book

Example

- Best ask = 50,050
 - You submit:
 - Limit buy @ 50,060 → fills immediately (crosses)
 - Limit buy @ 50,000 → rests in book
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Partial Fills

If your order is larger than available liquidity:

- You get filled **in chunks**
 - Remainder continues matching or rests
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4. Last Price

Last price = price of the **most recent trade**

Important:

- It is **not always reliable**
- One small trade can move it
- Easily manipulated with low volume

Used for:

- Candles (OHLC)
- PnL display
- Visual charts

Dangerous if used alone for:

- Liquidations
 - Stop-loss triggers
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5. Index Price

Index price = average price from **multiple spot exchanges**

Purpose:

- Represents the “real market value”
- Harder to manipulate
- Used as a reference anchor

📌 Example

Index price = average of:

- Binance spot BTC
 - Coinbase spot BTC
 - Kraken spot BTC
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6. Mark Price (Fair Price)

Mark price is a *derived price* used by futures exchanges.

Usually:

$$\text{Mark Price} \approx \text{Index Price} + \text{Funding Basis}$$

Why mark price exists

- Prevents fake wicks from liquidating traders
- Smooths volatility
- Stops stop-loss & liquidation hunting via tiny trades

📌 Used for:

- Liquidations
- Unrealized PnL
- Risk calculations

! You might see:

- Last price = 50,200
 - Mark price = 50,020
 - Liquidation uses **50,020**, not 50,200
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7. Funding Rate (Perpetual Futures)

Perpetual futures never expire, so **funding** keeps price aligned with spot.

How it works

- Paid **between traders**, not to the exchange
- Happens every X hours (e.g., 8h)

Logic:

- If perpetual price > spot:
 - **Longs pay shorts**
- If perpetual price < spot:
 - **Shorts pay longs**

Example

Funding = +0.01%

You are long \$100,000

You pay \$10 to shorts

Why funding matters

- Reflects **market positioning**
- High positive funding = crowded longs
- High negative funding = crowded shorts

8. Open Interest (OI)

Open interest = total number of **open contracts**

Important rule:

- OI increases only when **both sides open**
- OI decreases only when **both sides close**

Examples

Action	OI
Long opens + Short opens	↑
Long closes + Short closes	↓
Long closes + Short opens	↔

Why OI matters

- Measures **leverage in the system**
 - Rising OI + rising price → aggressive positioning
 - Falling OI + falling price → deleveraging
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9. How Liquidations Happen (Simplified)

- Liquidations are triggered using **mark price**
- Not last price
- Exchange force-closes position with **market orders**
- These market orders move price further → cascades

📌 This is why:

- Thin order books cause violent moves
 - Funding + OI + book depth together explain volatility
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10. Which Price Is Used for What?

Purpose	Price Used
Chart candles	Last price
Manual entry	Last price
Liquidation	Mark price
Unrealized PnL	Mark price
Fair value	Index price
Stops (configurable)	Last / Mark / Index

11. Putting It All Together (Mental Model)

Think of the exchange as:

- **Order book** → where liquidity lives
- **Market orders** → eat liquidity
- **Limit orders** → supply liquidity
- **Last price** → what just happened
- **Index price** → what should be happening

- **Mark price** → safety-adjusted truth
- **Funding rate** → pressure valve
- **Open interest** → leverage meter